

Day 8 — Cross-Cultural — Lo Shu and Dürer

Module: Magic Squares — Bhadrakaṇṭha · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare indian, chinese (lo shu), and european (dürer 1514) magic squares.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *siddhāṅka* (modern coinage) (IAST: *siddhāṅka* (modern coinage); lit. “the common sum of all rows/columns/diagonals”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The Lo Shu (Chinese).** Legendary: a turtle emerging from the River Lo bore the 3×3 magic square on its shell. Used in Chinese geomancy (*feng shui*). Earliest definite attestation: ~5th c. BCE (though legend places it ~2000 BCE).
2. **Indian — Khajuraho (10th c. CE).** As yesterday. Pan-diagonal 4×4 with “most-perfect” property.
3. **Indian — Nārāyaṇa Paṇḍita (1356 CE).** Systematic algorithms for any odd n .
4. **European — Dürer’s Melencolia I (1514).** A 4×4 with year-encoding.
5. **Class exercise.** Students draw a timeline (~–500 to ~+1600) marking each tradition’s main contributions. Then write one sentence: “*What does this timeline tell us about mathematical universality?*”

Critical-thinking caveat: “*Some claims of priority are unprovable. The TURTLE-LEGEND for Lo Shu is just that — a legend. We can only date things by extant manuscripts and inscriptions.*”

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Cross-Cultural — Lo Shu and Dürer.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).